

Working with Conditionals

Lab overview

A section of code that compares two pieces of information is called a *conditional statement*. You can use conditionals to create different paths through the program. Using comparative operators, you will write a program that makes decisions.

In this lab, you will:

- Use the `if` statement
 - Use the `else` statement
 - Use the `elif` statement
-

Estimated completion time

45 minutes

Accessing the AWS Cloud9 IDE

1. Start your lab environment by going to the top of these instructions and choosing **Start Lab**.

A **Start Lab** panel opens, displaying the lab status.

2. Wait until you see the message **Lab status ready**, and then close the **Start Lab** panel by choosing the **X**.

- At the top of these instructions, choose **AWS**.

The AWS Management Console opens in a new browser tab. The system automatically logs you in.

Note: If a new browser tab does not open, a banner or icon at the top of your browser typically indicates that your browser is preventing the site from opening pop-up windows. Choose the banner or icon, and choose **Allow pop ups**.

- In the AWS Management Console, choose **Services > Cloud9**. In the **Your environments** panel, locate the **reStart-python-cloud9** card, and choose **Open IDE**.

The AWS Cloud9 environment opens.

Note: If a pop-up window opens with the message *.c9/project.settings have been changed on disk*, choose **Discard** to ignore it. Likewise, if a dialog window prompts you to *Show third-party content*, choose **No** to decline.

Creating your Python exercise file

- From the menu bar, choose **File > New From Template > Python File**.

This action creates an untitled file.

- Delete the sample code provided from the template file.

- Choose **File > Save As...**, and provide a suitable name for the exercise file (for example, *conditionals.py*) and save it under the **/home/ec2-user/environment** directory.

Accessing the terminal session

- In your AWS Cloud9 IDE, choose the **+** icon and select **New Terminal**.

A terminal session opens.

- To display the present working directory, enter `pwd`. This command points to **/home/ec2-user/environment**.

- In this directory, you should also be able to locate the file you created in the previous section.

Exercise 1: Working with the if statement

In this exercise, you will edit a Python script to ship packages.

11. From the navigation pane of the IDE, choose the **.py** file that you created in the previous *Creating your Python exercise file* section.

12. Use the `input()` function to get information from the user:

```
userReply = input("Do you need to ship a package? (Enter yes or no) ")
```

13. Use the `if` statement to print a response.

The statements in an `if` statement are one tab indented from the `if` statement. In other programming languages, brackets are often used to indicate the start and end of a logic block, but Python uses spacing:

```
if userReply == "yes":  
    print("We can help you ship that package!")
```

Note: The `==` symbol is a comparative operator. It means *is equal to*.

14. Save and run the file.

15. At the prompt, enter `yes` and press ENTER.

16. Confirm that you see a response.

17. Run the file again.

18. At the prompt, enter `no` and press ENTER. Confirm that the program exits and nothing is displayed.

Exercise 2: Working with the else statement

To improve customer service, it would be nice to provide a reply even if the user doesn't want to ship a package. In this exercise, you will improve the Python script by using the `else` statement:

19. To handle the condition where the user doesn't want to ship a package, use the `else` statement:

```
else:  
    print("Please come back when you need to ship a package. Thank you.")
```

20. Save and run the file.

21. At the prompt, enter `no` and press ENTER.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

22. Confirm that you see a response.
 23. Run the file again.
 24. At the prompt, enter `yes` and press ENTER.
 25. Confirm that you see a response.
-

Exercise 3: Working with the `elif` statement

In this exercise, you will improve the Python script by offering the user additional services. When you have multiple conditions, you can use the `elif` statement, which is short for *else-if*.

Note: The `elif` statement always comes after an `if` statement and before the `else` statement.

26. In the Python script, enter the following code:

```
userReply = input("Would you like to buy stamps, buy an envelope, or make a copy? (Enter stamps, envelope, or copy) ")
if userReply == "stamps":
    print("We have many stamp designs to choose from.")
elif userReply == "envelope":
    print("We have many envelope sizes to choose from.")
elif userReply == "copy":
    copies = input("How many copies would you like? (Enter a number) ")
    print("Here are {} copies.".format(copies))
else:
    print("Thank you, please come again.")
```

27. Save and run the file.
28. At the prompt, enter `no` and press ENTER.
29. Confirm that you see a response.
30. At the prompt, enter `stamps` and press ENTER.
31. Confirm that you see a response.
32. Run the file again.
33. At the prompt, enter `yes` and press ENTER.
34. Confirm that you see a response.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

35. At the prompt, enter `envelope` and press ENTER.

36. Confirm that you see a response.
37. Run the file again.
38. At the prompt, enter `no` and press ENTER.
39. Confirm that you see a response.
40. At the prompt, enter `copy` and press ENTER.
41. Confirm that you see a response.
42. At the prompt, enter `2` and press ENTER.
43. Confirm that you see a response.

Note: The `if`, `elif`, and `else` statements allow only one path to run at a time. The program doesn't check the other statements after it finds a condition that is true.

As you can see, each time through the program had slightly different results. These differences demonstrate the power of conditionals.

Congratulations! You have written a Python script that uses `if`, `elif`, and `else` statements.

End Lab

🚩 Congratulations! You have completed the lab.

44. Choose **■ End Lab** at the top of this page, and then select Yes to confirm that you want to end the lab.

A panel indicates that *DELETE has been initiated... You may close this message box now.*

45. A message *Ended AWS Lab Successfully* is briefly displayed, indicating that the lab has ended.

Additional Resources

For more information about AWS Training and Certification, see <https://aws.amazon.com/training/> (<https://aws.amazon.com/training/>).

Your feedback is welcome and appreciated. If you would like to share any suggestions or corrections, please provide the details in our AWS Training and Certification Contact Form (<https://support.aws.amazon.com/#/contacts/aws-training>).

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